Script, Not Language: Where Multilingual Pretraining Stops Transferring in Sinhala–Tamil–English Ticket Triage

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***Abstract*—Sri Lankan bank customers write support tickets in Sinhala, Tamil, and English, and they write the two local languages in Latin characters as readily as in their native scripts. We present a parallel corpus of 13,077 banking tickets rendered in five tracks, 65,385 rows, derived from BANKING77 and extended with sentiment and priority labels, and use its parallel structure to locate where multilingual pretraining stops transferring. Because the Sinhala and romanized Sinhala tracks hold the same tickets in the same language and differ only in orthography, the comparison isolates script from language. A LaBSE encoder gains 8.3 points of Negative-F_1 over a character n-gram baseline on English and 8.2 on native Sinhala, and nothing measurable on either romanized track; as a difference-in-differences the gap is +0.0698 (95% CI $[0.014, 0.128]$, $p = 0.012$). The cause is tokenizer vocabulary coverage rather than sequence fragmentation: MuRIL maps 64.5% of Sinhala characters to `[UNK]` and loses 23.4 points on that track alone, while its subword fertility is the lowest we measured. Applying published scheduling rules to the resulting posteriors, the ordering rule moves queueing cost by 3.1× where the probability model moves it by 3%.**

***Index Terms*—multilingual NLP, code-mixed text, romanization, Sinhala, Tamil, support-ticket triage, tokenization, scheduling**

I. Introduction

A bank's support desk does not need a label. It needs to know which ticket to open next. Deployed triage systems predict an escalation risk, a component, or a severity tier and report accuracy against a held-out set [1]–[3], and stop there. A classifier that makes fewer mistakes can still order a queue worse than one that makes more, and no F_1 table will say so.

Sri Lankan retail banking adds a second problem that existing resources cover poorly. Customers write in Sinhala, Tamil, and English, and they write the two local languages in Latin characters at least as often as in the native abugidas. These romanized forms, Singlish and Tanglish, have no orthographic standard: one word admits several spellings, and English banking vocabulary is left untranslated mid-sentence. Multilingual encoders are pretrained on native-script web text. Whether their coverage of Sinhala and Tamil survives transliteration into Latin script has not, to our knowledge, been tested on parallel data.

Our corpus makes that test available. The same 13,077 tickets appear in five renderings, so a comparison between the Sinhala track and the Singlish track holds language, topic, label, and ticket identifier fixed and varies only the orthography. Any difference between them is a script effect, which lets us state the result as a difference-in-differences with a confidence interval rather than as two separate leaderboard columns.

The answer is that script coverage governs transfer. An encoder beats a character n-gram baseline by 8.3 points on English and 8.2 on native Sinhala, and by an amount indistinguishable from zero on both romanized tracks. The same shape recurs between two encoders and on a second task, and it has a concrete cause: a tokenizer that has never seen the Sinhala block emits `[UNK]` for 64.5% of its characters and collapses on that track alone. Subword fertility, the statistic tokenizer studies usually report, predicts none of it.

We then ask what the predictions are worth to a queue, applying published scheduling rules [4]–[6] unmodified to the posteriors and scoring each on the cost function its own theorem is stated over.

Contributions. (1) A parallel five-track corpus of Sri Lankan banking tickets with a frozen split and measured ceilings for its two generated label columns. (2) Evidence that the transfer boundary is orthographic rather than linguistic, tested on tickets differing only in script and reproduced across three comparisons. (3) A tokenizer-level mechanism, showing that out-of-vocabulary character rate predicts the downstream deficit where fertility does not. (4) A queue-level evaluation showing that the ordering rule dominates the classifier that feeds it.

II. Related Work

Ticket triage. Deployed systems predict a per-ticket label and report classification metrics: escalation risk over 2.5M IBM tickets [1], developer assignment for bug reports [2], and an industrial incident deployment [3]. None asks what the prediction is worth once tickets compete for a finite number of agents. BANKING77 [7] supplies our English track and its human intent labels.

Multilingual encoders. XLM-R [8] and LaBSE [9] transfer across many languages; MuRIL [10] and IndicBERT [11] narrow the target set to Indian languages, TwHIN-BERT [12] to social media, and mmBERT [13] widens it again. Transfer is almost always measured on native-script data, because that

TABLE I

CORPUS COMPOSITION. ALL FIVE TRACKS HOLD THE SAME 13,077 TICKETS AND THE SAME 77 INTENTS.

Track	Provenance	Script	% Latin
english	BANKING77, unmodified	Latin	100.0
sinhala	MT, hand-corrected	Sinhala	40.4
singlish	rule-based romanization	Latin	100.0
tamil	machine-translated	Tamil	1.3
tamilish	MT romanization	Latin	100.0

is what pretraining corpora and benchmarks contain, so the distinction we draw in Section VI is one these benchmarks cannot make. CANINE [14] removes the failure mode architecturally by operating on Unicode code points.

Romanized South Asian text. Chakravarthi et al. [15] released human-typed Tanglish, the reference point that makes our synthetic romanization an optimistic bound. Sinhala resources are thinner [16]: work covers back-transliteration [17], Sinhala-specific pretraining [18], and script effects in Sri Lankan text [19]. Parallel banking corpora across these renderings have not been released.

Labels and scheduling. Two of our label columns are model-generated, so we report measured ceilings, following the treatment of benchmark label error in [20]. For ordering we apply the $c\mu$ rule [4], its generalisation to convex cost [5], and Argon and Ziya's result that ordering by a posterior beats ordering by its argmax [6]. Because a ticket carries three labels, we compare binary relevance against classifier chains [21], probabilistic chains [22], and stacking [23].

III. THE CORPUS

A. Construction

The corpus holds 13,077 retail-banking support tickets in five parallel renderings, 65,385 rows. The English track is BANKING77 [7] taken unmodified, so its 77-way intent column is human gold. The other four derive from it: Sinhala and Tamil by machine translation, Singlish by rule-based romanization of the Sinhala track, and Tanglish by machine-translated romanization (Table I).

The tracks are aligned row for row. A ticket keeps its identifier and all three labels across all five, so any pair of tracks is a matched sample. That property carries the experiment in Section VI and is why the corpus was built this way.

B. Split and labels

We draw one split at the ticket level and fan it across all five tracks, so a ticket seen in training never appears in test in any rendering. Splitting per track would place the Sinhala and Singlish versions of one ticket on opposite sides of the boundary, a leak specific to parallel corpora. The frozen split is 8,500 / 1,498 / 3,079 tickets, or 42,500 / 7,490 / 15,395 rows.

Only intent is human gold. Sentiment and priority were generated by prompting a large language model over the English text and propagated to the other tracks, so both carry label noise [20]. Reporting scores without reporting what the labels are worth would overstate every result here. Measured against a 500-ticket human-annotated gold set, sentiment agreement reaches 0.7931 Negative-F_1 (κ = 0.78) and priority 0.7722 macro-F_1 (κ = 0.64). Every score below should be read against those ceilings.

Two distributional facts fix our metrics. Sentiment is 93.7% Neutral, so accuracy is uninformative and we report Negative-F_1 throughout. Priority is 53.5 / 36.6 / 9.8 across Low, Medium, and High, so the operationally important class is the rarest.

C. Two defects in our own corpus

The Tanglish test rendering does not match its training rendering. Tanglish intent scores 69.3 macro-F_1 on test against 91.7 on development, where no other track moves more than three points. The cause is vocabulary, not generalisation: 60.3% of Tanglish test tokens are unseen in training, against 20–30% elsewhere. Singlish is generated by a deterministic rule and emits one spelling per word; Tanglish is machine-translated and emits several, needing 7,250 training vocabulary types against Singlish's 3,293, with only 39.7% of its test types appearing in its own training data. We state this here rather than only in results, because a reviewer who meets it later will not trust this section. Every Tanglish test number below is a lower bound and the track is excluded from every claim the paper rests on.

The two native tracks were not built to one standard. The translation prompt instructed that English banking vocabulary be left untranslated, which is how customers write. Over a 31-term banking lexicon, the Sinhala track retains 64.6% of those terms in English and Tamil 7.6%; the Latin fractions in Table I say the same. Only the Sinhala track implements the code-mixing the documentation describes. Two consequences bind Section VI: register claims hold for one track of two, and the Tamil–Tanglish pair is not a second measurement of the Sinhala–Singlish pair, because the two pairs do not share a baseline.

IV. METHODOLOGY

A. Tasks, models, and protocol

Each ticket carries three labels and we treat them as three tasks over the same input: 77-way intent and 3-way priority scored with macro-F_1, and binary sentiment scored with Negative-F_1, the minority class alone.

Four families are scored on the same rows. **Classical:** word (1–2 gram) and character (`char_wb` 3–5 gram) TF-IDF vectors feeding a linear SVM, logistic regression, SGD, and complement naive Bayes. Character n-grams make this a serious baseline on romanized text, because they degrade gracefully when spelling is unstable. **Encoders:** LaBSE [9], XLM-R [8], mmBERT [13], MuRIL [10], IndicBERT [11], and TwHIN-BERT [12] with a linear head, AdamW at 2×10^{-5}, batch 32, sequence length set per model at its own 99th-percentile tokenised length. **Decoders:** Gemma 3 at 270M and

1B [24] with LoRA [25] at rank 8 over all projections. **Frozen probes:** logistic regression over frozen representations. Every model also runs under three class-imbalance treatments, and tables report the best arm.

Development is fitted on training data alone and used for all selection; test is fitted on training plus development and scored once. No table mixes the two. Every record is stamped with the split hash, label version, seed, and epoch-selection rule.

B. Inference

Three of our claims turn on differences small enough that the test procedure decides the answer. Differences between systems use a paired bootstrap over ticket identifiers, 1,000 resamples [26], with exact McNemar tests [27] alongside. The two can disagree, because McNemar tests accuracy and our headline sentiment metric is not accuracy; on three of 48 comparisons they do, and in two of those the disagreement would have asserted a cross-lingual effect absent from the reported metric.

Script contrasts are tested as differences of differences. Reporting an effect as significant on one track and not another is not a test that the two differ [28], and our parallel corpus makes the correct test available directly on paired identifiers.

C. Budget and a defect we found

An earlier version of our training loop selected the reported epoch by best score on the evaluation set. On development that is model selection; on test it makes the figure a maximum over n draws rather than a held-out estimate. We fixed it so test runs report their final epoch. Selection can only inflate a score where it had something to choose, so of the affected records 15 are inert exactly and two carry real bias, both on minor models we exclude.

Across 25 fine-tuned test records, 23 end on their best epoch, which alone reads as an undertrained roster. The per-epoch curves settle it: sentiment converges by three epochs (last gain 0.0011), priority sits on a plateau, and intent is still climbing at six, so every intent score here is a lower bound. Three further models are undertrained by this test and dropped from comparative claims.

V. Roster Results

Table II reports the roster on the frozen test set, fitted on training plus development and scored once.

Three readings are supported and a fourth is not.

The encoder advantage is real on every task and largest on sentiment. LaBSE beats the linear SVM by +0.0485 Negative-F_1 (95% CI [0.026, 0.072], McNemar $p = 1.4\times10^{-7}$) and by +0.0526 macro-F_1 on intent (CI [0.048, 0.058], $p = 3.2\times10^{-87}$). Section VI shows this pooled figure hides where the gain lives.

Sentiment is a tie at the top and we name no winner. LaBSE, Gemma 3 1B, and two multitask variants span 0.0096, while a single seed repeat of LaBSE moved it by 0.0158 at matched budget, larger than the spread it would decide. The defensible statement is that every encoder and decoder lands at 0.70 ± 0.01 against a classical floor of 0.665, and that the best reaches 90.0% of what the labels themselves support.

A 270M decoder buys nothing over character n-grams on intent, scoring 0.8414 against 0.8308 at far higher training cost, with both five points below LaBSE. At this corpus size, parameter count is not what separates systems.

What the table does not support is a decoder scaling claim as originally run, since the 1B model fell back to batch 8 after an out-of-memory failure. Re-running the 270M model at batch 8 leaves parameter count as the only difference and gives +0.0221.

TABLE II
Pooled test scores, best imbalance arm per model. Intent and priority are macro-F_1, sentiment Negative-F_1. Dashes mark cells not yet run.

Model	Intent	Sentiment	Priority
LaBSE	**0.8835**	**0.7138**	0.8900
XLM-R base	0.8801	0.7007	0.8872
mmBERT	0.8680	0.7000	0.8887
MuRIL	0.8467	0.6790	0.8717
Gemma 3 1B	0.8635	0.7126	0.8898
Gemma 3 270M	0.8414	–	–
TF-IDF + linear SVM	0.8308	0.6653	0.8734
TF-IDF + logistic	0.8189	0.6383	0.8706
TF-IDF + complement NB	0.6792	0.4968	0.8422
LaBSE frozen probe	0.7822	–	0.8015
Label ceiling	n/a	*0.7931*	*0.7722*

TABLE III
LaBSE minus TF-IDF+SVM, sentiment Negative-F_1 on test, paired bootstrap within each track (n = 3079 tickets per track). The last two rows are differences of differences between paired tracks.

Track	Script	Gain	p
english	Latin (seen)	+0.0834	< 0.001
sinhala	Sinhala (seen)	+0.0822	0.004
tamil	Tamil (seen)	+0.0357	0.144
singlish	Latin (unseen form)	+0.0124	0.712
tamilish	Latin (unseen form)	+0.0225	0.490
sinhala − singlish		**+0.0698**	**0.012**
tamil − tamilish		+0.0132	0.762

VI. The Transfer Boundary Is Orthographic

A. Where the advantage lives

The pooled sentiment gain of +0.0485 is not spread evenly across the five tracks (Table III). Every point of it is earned on text written in a script the encoder was pretrained on. On the two romanized tracks the encoder and a bag of character n-grams are indistinguishable.

The decisive row is the second from last. The Sinhala and Singlish tracks hold the same tickets, in the same language, with the same labels and identifiers, differing only in whether the text is written in the Sinhala abugida or transliterated into Latin characters. A difference in gain between them cannot be a language, topic, or sampling effect. Tested directly it is

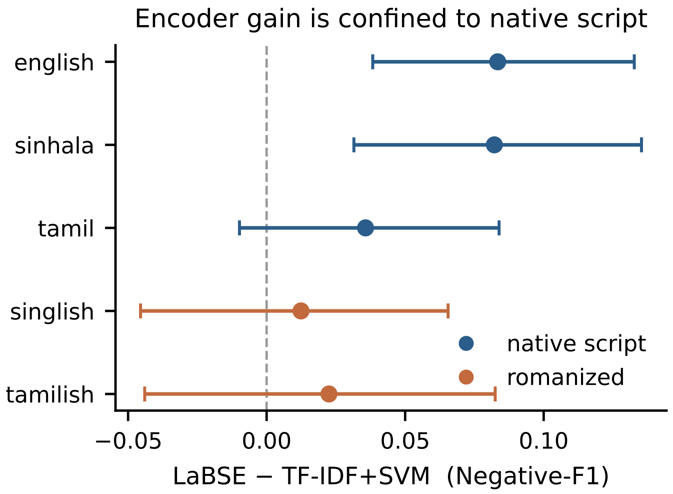

Fig. 1. Encoder gain over the classical champion by track, 95% bootstrap intervals. The gain is significant only where the script was in pretraining.

+0.0698 with a 95% interval of $[0.014, 0.128]$, $p = 0.012$. Fig. 1 shows the same decomposition with intervals.

The Tamil pair does not replicate ($p = 0.762$) and we do not claim it does. Section III gives the reason to expect it will not: the two pairs differ in their own English content, 64.6% loanword retention against 7.6%, so they do not share a baseline. The finding holds for Sinhala and is open for Tamil.

B. The same shape between two encoders

If the boundary is orthographic it should appear wherever two systems differ in script coverage, not only between an encoder and n-grams. LaBSE and mmBERT are tied on intent development ($p = 0.944$) and separate on test by +0.0155 macro-F_1. That gap is confined to the two non-Latin scripts: sinhala +0.0196 ($p = 1.8 \times 10^{-6}$) and tamil +0.0178 ($p = 2.8 \times 10^{-5}$), against english +0.0037 and singlish +0.0022, both non-significant. English and Singlish are both Latin and both show nothing. The result survives dropping the defective Tanglish track (+0.0106, $p = 7.1 \times 10^{-7}$).

C. The mechanism is vocabulary coverage

Tokenizer studies usually report subword fertility, the pieces produced per word, on the reasoning that a tokenizer shattering a language will encode it poorly. On our data fertility predicts nothing and out-of-vocabulary character rate predicts the deficit.

MuRIL is the clearest case. Its tokenizer maps 64.5% of Sinhala characters to `[UNK]` and 0.0% of every other track, including both romanized ones. Its Sinhala sentiment score falls 23.4 points below its own English score, while its Sinhala fertility of 1.18 is the *lowest* of any model–track pair we measured. It is not fragmenting Sinhala. It is deleting it. Across all twenty pairs the deficit correlates with `[UNK]` rate at $\rho = -0.635$ and with fertility at $\rho = +0.216$, the wrong sign for the fragmentation hypothesis altogether.

This explains why an Indic-targeted encoder underperforms a general multilingual one on Sri Lankan text. MuRIL covers Indian languages and the Sinhala Unicode block is not among them. The model does not transfer poorly to Sinhala. It never receives Sinhala.

The hole recurs on a second task and shrinks with the label space. MuRIL scores 0.8072 macro-F_1 on native Sinhala priority against 0.9095 on English, a 10.2-point deficit on the same tickets whose intent deficit is 20.0 points. The input destruction is identical; a three-way task tolerates roughly twice as much of it as a 77-way one. A coarse triage tier can survive a tokenizer that a fine-grained intent router cannot.

TABLE IV
PUBLISHED POLICIES AT $\rho = 1.05$ UNDER THE BEST LABEL MODEL, 200 SEEDS. COST COLUMNS ARE MEAN DELAY COST, LOWER IS BETTER. EACH RULE WINS ON THE OBJECTIVE ITS OWN THEOREM OPTIMIZES AND IS MID-TABLE ON THE OTHER.

Policy	Linear	Convex	High wait (min)
$c\mu$ / HSF [4]	**0.121**	0.137	1.77
Generalised $c\mu$ [5]	0.149	**0.064**	3.29
Earliest due date	0.146	0.111	4.78
Accumulating priority	0.197	0.099	9.34
Static tier (argmax)	0.161	0.163	6.70
First come, first served	0.381	0.625	52.46

VII. From Labels to a Queue

A desk with five agents and a backlog consumes an order, not labels, and two systems with equal macro-F_1 can produce very different orders. We therefore evaluate the predictions at the level they are used: rather than design a scoring function, we apply published scheduling rules unmodified and score each on the cost objective its own theorem is stated over. The simulator is an M/G/5 queue with log-normal service times (mean 8 minutes, $\sigma = 0.75$) at utilisation ρ, 200 seeds, with service windows of 30, 120, and 480 minutes for High, Medium, and Low.

The three labels are not three independent signals. On the English training track, intent carries 0.7178 nats of mutual information with priority against priority's entropy of 0.9333 nats, 76.9% of what there is to know, so an additive score over an intent head and a priority head adds two estimates of one quantity and its fitted weight measures collinearity. Sentiment is the mirror image: 0.0172 nats, of which 86.6% survives conditioning on intent. It knows little, and what it knows is not already known. Neither fact is visible in a weight, which argues for modelling the joint distribution.

Comparing five ways of forming a joint priority posterior, a log opinion pool reaches 0.2683 log-loss against binary relevance's 0.3697, a 27% reduction, while macro-F_1 moves across a range of 0.024 with no consistent ordering. Modelling label dependence improves calibration and not argmax accuracy, which is exactly the quantity a scheduling index consumes and an accuracy table cannot report. Classifier chains [21], probabilistic chains [22], and stacking [23] fall between.

Each rule wins on its own objective. A bake-off ranking policies on some other quantity is not testing them. The $c\mu$ rule and Argon and Ziya's Theorem 3 are stated over linear delay cost and the generalised rule over convex cost; Table IV measures both. The $c\mu$ rule sweeps linear cost and sits mid-table on convex, and the generalised rule does the reverse, so the choice between them is a choice of cost model rather than an empirical contest.

Argon and Ziya's result holds when tested on the objective it is stated over: against the same label model the posterior index beats the static tier every time, 0.121–0.125 against 0.161–0.168, ranges that do not overlap, a 25% cost reduction. Scored on relative tardiness, which no cited theorem mentions, the same comparison is a tie and the theorem appears to fail.

The ordering rule dominates the probability model. Under linear cost the five label models span 3% while the policies span a factor of 3.1. Practitioners choosing where to spend effort should read that ordering. Under convex cost the label model earns more, spanning 42%, so dependence modelling pays once the cost is convex. At $\rho = 0.85$ nothing separates: linear cost runs 0.019–0.022 across every combination. With no backlog there is nothing to order, and every claim here is an overload claim.

VIII. Discussion and Limitations

A. What to deploy

Choose an encoder by script coverage before benchmark rank. An Indic-targeted model is the intuitive choice for Sinhala and Tamil, and MuRIL is the weakest neural system we measured on all three tasks, losing to character n-grams on priority. The check costs a minute and no training: run the tokenizer over the target text and count the `[UNK]` rate.

Do not assume a native-script gain survives romanization. On our romanized tracks a linear model over character n-grams matches a multilingual encoder, so deployments evaluated only on native-script data overestimate what the encoder buys, in a direction that disadvantages exactly the customers who write in romanized form.

Spend on the ordering rule before the classifier. Changing which published rule consumes the posterior moves cost by a factor of three; changing the probability model beneath it moves cost by 3%.

B. Limitations

Each limitation below is measured rather than hedged. **The romanized tracks are synthetic**, generated from the native tracks rather than typed by people, so they lack the spelling variation of real romanized text [15]. Real input is harder, which makes our romanized scores an optimistic bound and the script gap a conservative estimate. **The Tanglish test rendering is defective** (Section III) and is excluded from every claim the paper rests on; the one finding it appears to strengthen survives its removal. **Two label columns are model-generated**, and we report their ceilings beside every score. **The sentiment ceiling rests on one annotator and 31 Negative tickets**, giving an interval of $[0.667, 0.896]$ that the 90.0%-of-ceiling claim inherits. **Intervals are within-fit**: they resample the test set, not the training run, and one seed repeat moved the sentiment headline by 0.0158, so differences below roughly 0.02 on sentiment should not be read as system differences. **Intent scores are lower bounds**, still improving at the six-epoch budget. **The queue is simulated with assumed parameters**: arrival rate, service distribution, agent count, and service windows are assumptions, mitigated by the load sweep but the largest threat to Section VII. **The Tamil contrast is not established**, and **the vocabulary mechanism is identified, not isolated**; a character-level control at convergence [14] would close the latter.

IX. Conclusion

We presented a parallel five-track corpus of 13,077 Sri Lankan banking support tickets and used its structure to locate where multilingual pretraining stops transferring. Because the same tickets appear in native script and in transliteration, the comparison isolates orthography from language, and the encoder's advantage over a character n-gram baseline lives entirely on scripts seen during pretraining: +0.0698 as a difference of differences, with the same shape reproduced between two encoders and on a second task. The mechanism is tokenizer vocabulary coverage rather than sequence fragmentation, which makes it diagnosable before training and unfixable by fine-tuning.

Applying published scheduling rules to the resulting posteriors, the ordering rule moves queueing cost by a factor of three where the probability model moves it by 3%, and modelling dependence among the three labels improves calibration by 27% without improving accuracy. A study reporting only macro-F_1 would have missed both.

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Portions of the prose in Sections I–IX were drafted with the assistance of a large language model (Anthropic Claude, Opus 5) used at the level of drafting and copy-editing from author-supplied experimental results, outlines, and notes. The system generated no experimental result, number, table, or figure. All experiments, analyses, and conclusions are the authors' own; every reported value is produced by a script in the released code and was verified against its generating artifact. The authors reviewed and edited all such text and take full responsibility for the content of this article.

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